StockSense AI

Sales forecasts and stock suggestions for a fictional mini market

Store 1 5 products 7-day forecast

The weekly stock decision

How much should the manager order before next week’s sales are known?

Too little stock

Missed sales

Too much stock

Money tied up in inventory

A small, clear prototype

1

fictional store

5

anonymous products

7

forecast days

The app turns these forecasts and entered stock values into an order suggestion.

The sales data

913,000

rows in the source file

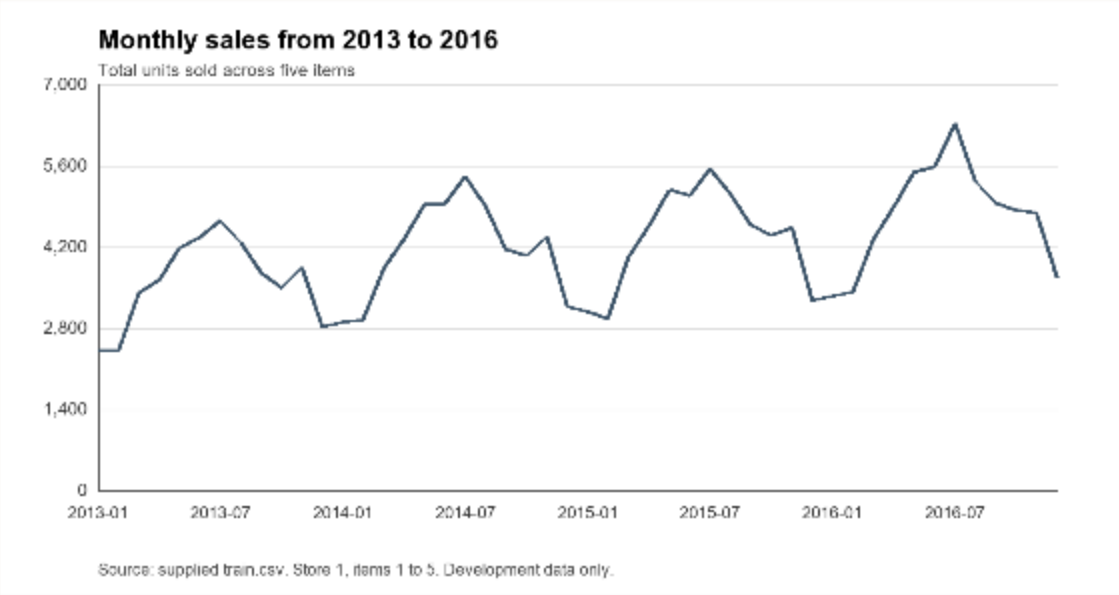
9,130

rows in our selected store and items

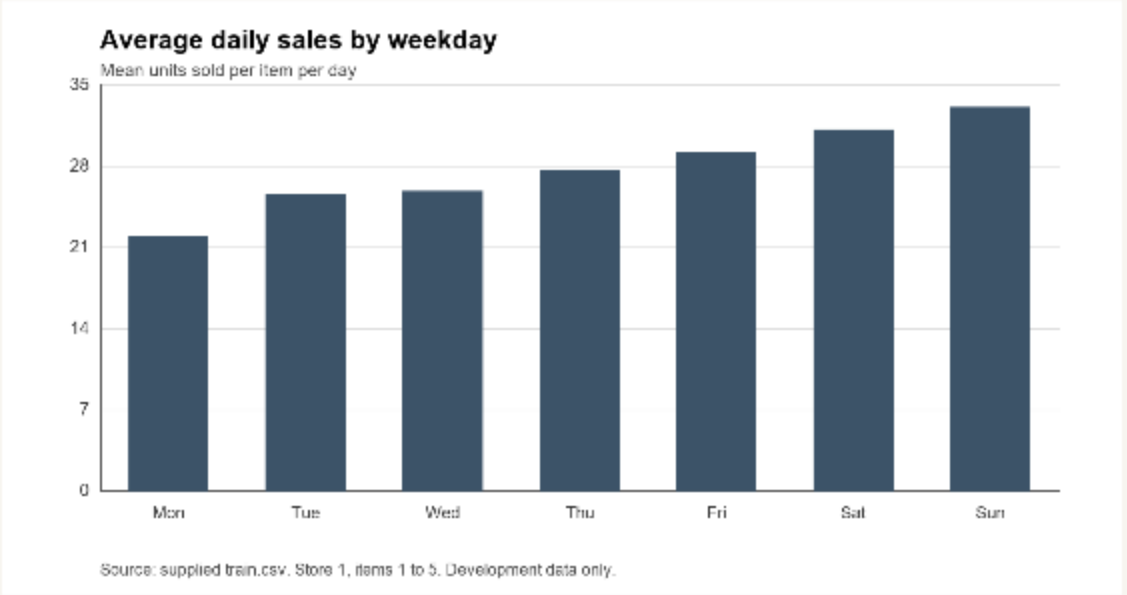
Daily sales from 2013 to 2017

No product names, stock levels, prices or supplier records are supplied.

Sales patterns before testing



Monthly sales totals



Average by weekday

Time split and methods

2013–15

Fit the models

2016

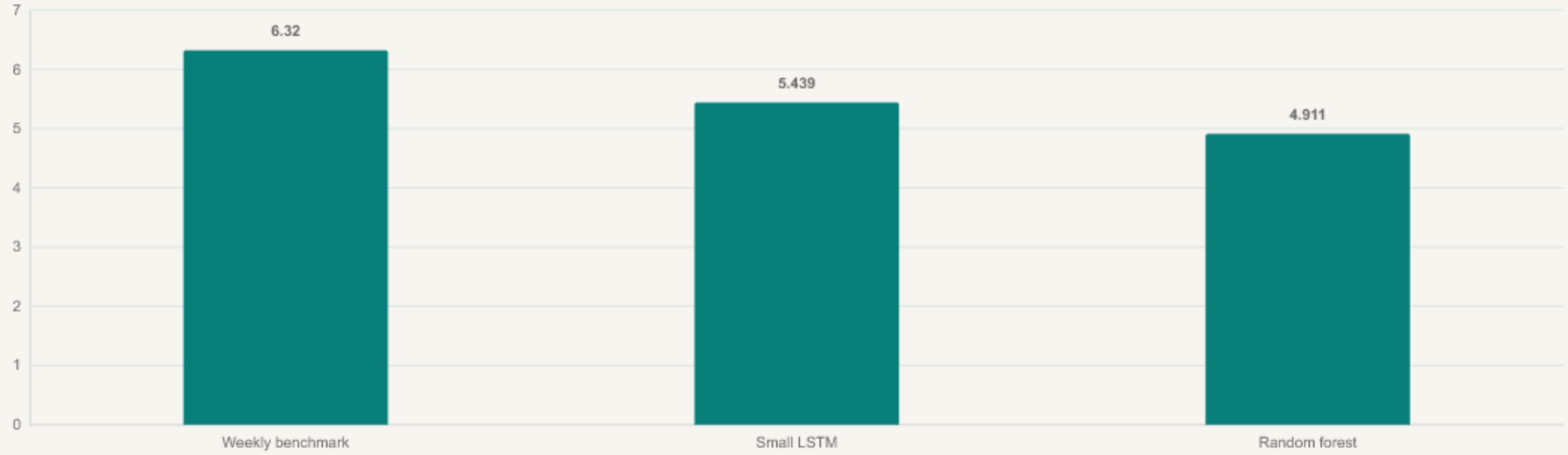
Select by daily MAE

2017

Final test

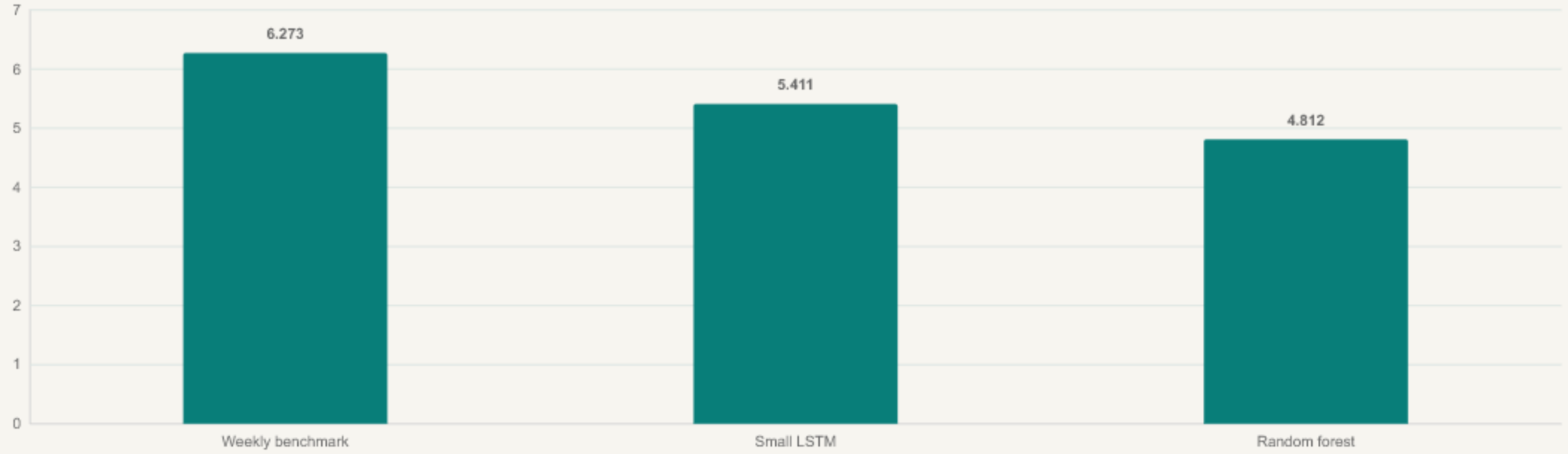
Compared: last-week repeat, random forest and a small LSTM

Random forest won validation



Daily mean absolute error (units)

The result held on the final test



23.3% lower daily error than the weekly benchmark

LSTM had the lowest weekly-total error (17.343 units).

A small LSTM comparison

28

past sales days

16

LSTM units

7

predicted days

It had better weekly totals overall, while random forest had lower daily error.

How the order number is calculated

77.52

forecast units during coverage

+ 10 buffer = 87.52 target

50 available, 0 outstanding

Suggest ordering 38 units

Ask StockSense

“What will Product 1 sell next week?”

The assistant identifies the request type, then reads the saved forecast.

5 supported request types

Forecast, order, recent sales, model results and help

Prototype checks

79

automated tests passed

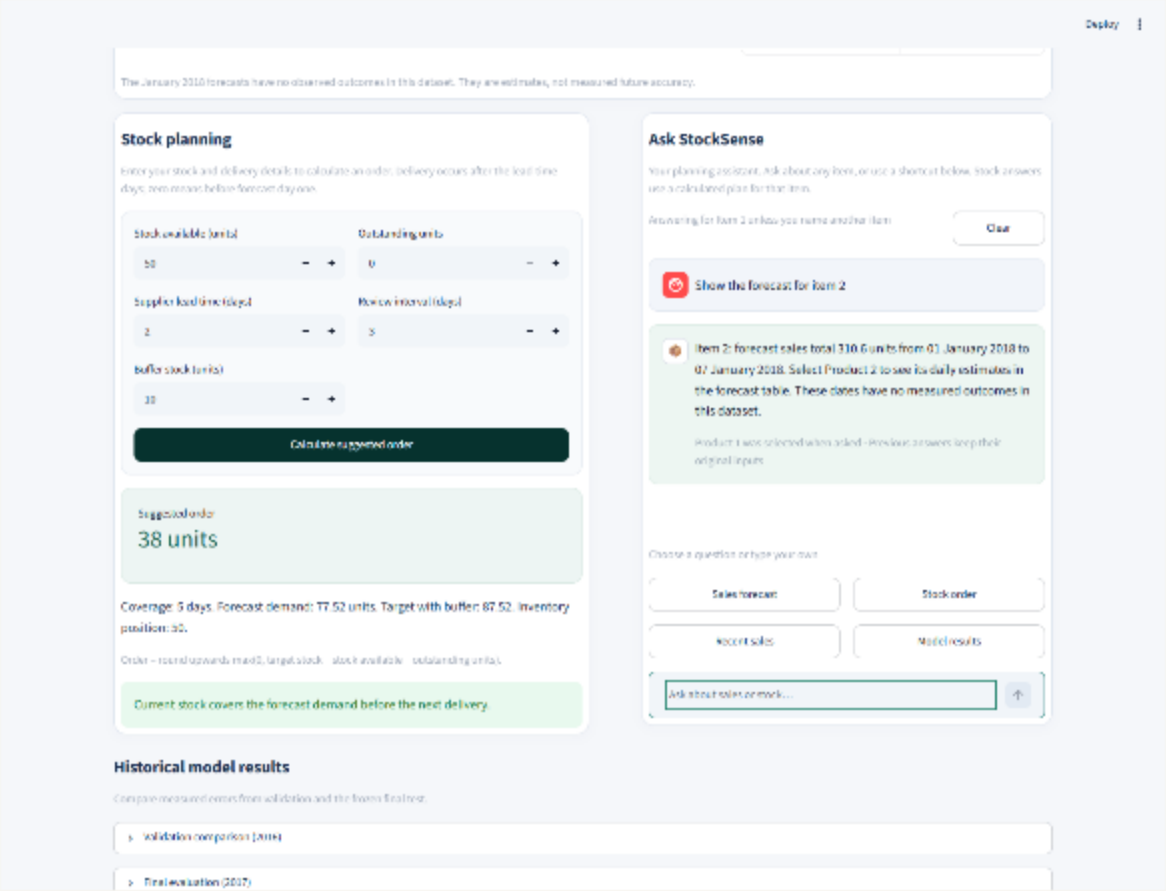
24/25

supported requests recognised

8/8

unrelated requests rejected

Live prototype walkthrough



01 Select a product

02 View seven days

03 Enter stock

04 Ask a question

What we can conclude

Random forest had the lowest daily test error.

The app turns forecasts into a clear order calculation.

Real savings and January 2018 accuracy remain unmeasured.

Questions